Scorer-Induced Partial Identification of Benchmark Comparisons: An Audit-Design Approach

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Abstract

Benchmark leaderboards report model accuracy as if it were an observed quantity. It is the output of an automated verifier applied to free-text generations, and that verifier can disagree with the correctness a human reader would assign. We formalize this as a partial-identification problem: given a benchmark's observed accuracy and an *audited* estimate of the verifier's false-credit and false-miss rates, we derive the sharp set of true accuracy values consistent with the data, adapting classical misclassification-bounds machinery [Manski and Horowitz, 1995] to benchmark comparison. On MATH-Hard, using a real human audit of the standard verifier, we show that **audit-estimation uncertainty—how little the verifier's own error rate has been pinned down—exceeds sampling uncertainty for the majority of model comparisons that remain identifiable at all** (median identification-to-sampling width ratio $6.10\times$, 100% dominant among 120 well-defined pairwise comparisons out of 136 candidates among 17 qualifying models; a self-consistency check using the audited comparator's own accuracy rather than the field's reported metric gives $5.74\times$, also 100% dominant; minimum ratio across all 120 valid pairs is $2.15\times$, so the result is not driven by a few extreme comparisons). We formalize the resulting audit-budget allocation problem, prove the classical [Neyman, 1934] solution, and show empirically that its naive plug-in form is *dominated by uniform allocation* on our own sparse real audit data—a documented failure mode with a tested correction. A secondary finding: MATH-Hard's evaluation harness computes accuracy via two live, independently-implemented extraction procedures that disagree on 3.63% of real responses (95% CI [3.39%, 3.88%], $n = 22{,}508$); this scorer-choice dependence does not overturn the dominance finding, including under an adversarial worst-case bound, but is itself a distinct, measurable source of benchmark uncertainty. [PENDING — human audit: the verifier's other live scorer has not itself been human-audited; its own α, β and the fully self-consistent identification result built on them await a preregistered 458-row audit already in the field].

1 Introduction

Benchmark leaderboards report accuracy as if it were observed. It is not. A model's reported score is the output of a fixed, deterministic verifier applied to free-text generations—a string comparison, a regex over a boxed answer, a symbolic equivalence check—and that verifier can be wrong. When it is,

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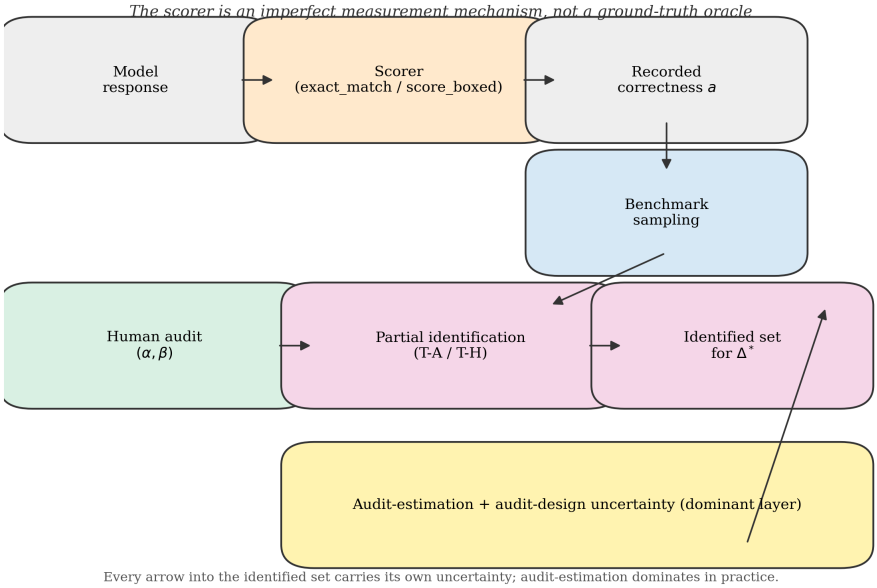


Figure 1: The scorer is an imperfect measurement mechanism, not a ground-truth oracle. Every arrow into the identified set carries its own uncertainty; audit-estimation dominates in practice (Section 7.2).

the number on the leaderboard is not the model's accuracy; it is the model's accuracy contaminated by the verifier's own error rate, in a direction and magnitude nobody has measured. Figure 1 sketches the resulting measurement pipeline and where each source of uncertainty enters it.

This paper asks what a benchmark comparison is still entitled to claim once that contamination is taken seriously. We audit MATH-Hard's standard comparator on real model responses, find it disagrees with human judgment at a small but nonzero rate, and derive the set of true accuracy values consistent with what was actually observed—not a point estimate corrected by a guessed error rate, but a *partially identified set*, sharp under the weakest assumption the audit data will bear. The central empirical finding is not that the verifier is wrong often. It is that **how much a benchmark's verifier has been audited turns out to matter more than how good the benchmark's sample size is**: audit-estimation uncertainty, driven by how few human-labeled items exist to pin down the verifier's error rate, exceeds the field's already-reported sampling uncertainty for the majority of real model comparisons we can identify at all.

That finding immediately raises a design question the literature has not answered: if verifier auditing is the binding constraint, how should a fixed human-labeling budget be spent? We formalize this as a stratified-sampling allocation problem, derive the classical [Neyman, 1934] solution, and then show—on this paper's own real, sparse audit data—that the classical formula's own assumption (a decent pilot estimate) fails in exactly the regime most benchmarks are actually audited in: when most audited strata have observed zero error events, naive plug-in allocation concentrates budget on whichever cell happened to draw a noisy nonzero rate, and is dominated by simply spreading the budget evenly. A stress-tested, sparse-data-aware allocation rule is this paper's most directly usable output.

A third thread emerged from building an adversarial test suite for the verifier: MATH-Hard's evaluation harness computes accuracy two different ways from the same generations—one via LaTeX-aware symbolic parsing (`exact_match`, via `math_verify`), one via locating the last `\boxed{}` span (`score_boxed`)—and these two, both live, both in active use, disagree on 3.63% of real responses (95% CI [3.39%, 3.88%], full population, 17 models $\times$ 7 subjects). Every human audit this paper reports was conducted against `score_boxed`; the field's reported accuracy comes from `exact_match`.

We show this mismatch does not overturn the audit-estimation-dominance finding—tested directly, not assumed, including under a deliberately adversarial worst-case bound on the unaudited scorer's error rate—but it establishes a second, independent source of benchmark uncertainty: *which* automated criterion is used to score a benchmark is itself a measurable, sometimes consequential, choice. Independent, production-scale confirmation that this general problem is real (not specific to our framing) is discussed in Section 2.

Contributions. (1) A sharp partial-identification treatment of benchmark accuracy under an audited, imperfect verifier, adapting classical misclassification-bounds machinery [Manski and Horowitz, 1995, Molinari, 2008] to the benchmark-comparison setting—the adaptation, not the underlying mathematics, is the contribution. (2) An empirical demonstration, on real data from a widely-used benchmark, that audit-estimation uncertainty dominates sampling uncertainty for the majority of identifiable model comparisons. (3) A stratified audit-design methodology (Section 5), including a documented failure mode of the textbook-optimal allocation rule under sparse real data and a corrected, tested alternative. (4) A secondary, honestly-scoped finding that automated-scorer choice is itself a measurable, non-dominant but non-negligible source of benchmark uncertainty, with a worst-case sensitivity bound rather than an assumed-safe headline. Sections 3–4 develop the identification theory; Section 5 the audit-design problem; Sections 6–7 the empirical results, including the scorer-choice and worst-case analyses; Section 8 limitations.

2 Related Work

Partial identification and misclassification. The identification strategy here is a direct adaptation of the classical misclassification-bounds literature [Manski and Horowitz, 1995, Molinari, 2008]—sensitivity/specificity-style calibration constants bounding a corrupted binary outcome. We claim no new mathematics in this layer (class A/B throughout, see the supplementary novelty audit, Appendix A.9); the contribution is applying it to benchmark comparison specifically, with an audit-design layer classical misclassification papers do not address.

Concurrent, independent work. Chen et al. [2026] study partial identification of a *prevalence* from panels of LLM *raters* whose errors are correlated because the raters share training lineage—the same classical toolkit, a structurally different problem (their LLMs are the noisy raters; ours are the *rated* models, scored by one fixed, non-LLM, deterministic verifier). No overlap in estimand, no overlap in headline results; cited as the closest neighbor, not a competing claim.

Concurrent work by the same author, sharing empirical infrastructure, distinct in estimand and contribution. Pall [2026] (under review, non-archival at the time of writing) formalizes reported accuracy as $\hat{A} - \bar{\varphi} \leq A^* \leq \hat{A} + U/n + \psi$, bounded primarily by a scorer's own discarded parse-failure count U, and reports that 90.2–94.4% of 378 pairwise orderings on a 28-model MATH-Hard roster are not identified. That paper and this one draw on the *same* 400-item human audit (same MATH-Hard snapshot, same 28-model base roster) as a shared empirical resource, disclosed here explicitly rather than left for a reader to discover independently. The two are not the same result under different notation: this paper's estimand is a full two-parameter false-credit/false-miss box (α, β), not conditioned on parse failure specifically, and its headline statistic (median identification-to-sampling width ratio) is not the other paper's (% of orderings unidentified). Two contributions here have no counterpart there at all—the audit-budget-allocation theory and failure mode (Section 5), and the scorer-choice-dependence finding between two named, live production scorers (Section 7.4)—while that paper's cross-harness census (`lm-evaluation-harness`, OpenCompass) and learned-extractor comparison have no counterpart here. See 'VENUE_AND_OVERLAP_AUDIT.md' in the code release for the full side-by-side comparison and the policy analysis this disclosure is based on.

Benchmark and verifier reliability. VerifyBench [Li et al., 2025] and CompassVerifier [Liu et al., 2025] study *which verifier is more accurate*—a different question from ours ("given a verifier's audited error rate, what can still be concluded about a comparison"), complementary rather than

overlapping. A body of work independently documents answer-format/extraction dependence as an open problem in LLM evaluation generally; we do not claim to have discovered that extraction is fragile, only to formalize its statistical consequence for benchmark comparison.

Independent, high-value external corroboration. Hugging Face's own Open LLM Leaderboard team documented, in a public postmortem [Kydlicek et al., 2025], the same family of comparator limitations this paper's own adversarial suite independently found for `score_boxed`'s SymPy-based comparison step—no rounding-tolerance support, no numerical-equivalence support, no variable-assignment support, no matrix- or set-equivalence support—and switched the leaderboard's own scoring to `math-verify`, the same library this paper's `exact_match` field uses, after finding the old comparator's failures were severe enough to more-than-double some model families' scores (Qwen) and nearly triple others' (DeepSeek, whose `\boxed{}`-wrapped answers the old comparator could not extract at all). We do not claim `score_boxed`'s exact extraction path here is byte-identical to the specific historical scorer HF retired—only that an independent, production-scale, industry audit of the same comparator family found failures large enough to force a platform-wide rescoring of 3,751 models, which is direct, external, non-self-reported evidence that scorer choice is a real, consequential, previously-underscrutinized problem for exactly this benchmark family—not a phenomenon manufactured by this paper's own framing. It also means our `exact_match` accuracy input is not an arbitrary alternative scorer: it is the field's own currently-endorsed replacement metric.

Audit design. The stratified-allocation solution (Section 5) is Neyman [1934], unmodified—cited, not claimed novel. The contribution is a documented, tested failure mode of the plug-in form under the specific sparse-event regime real benchmark audits actually operate in, and a corrected allocation.

Verification bias. The stratified-on-scorer-outcome audit design is checked against, and shown formally distinct from, the classical partial-verification-bias problem in diagnostic testing [Begg and Greenes, 1983]—our sampling probabilities are known by design, not an unknown behavioral process, which is the condition under which Horvitz–Thompson estimation [Horvitz and Thompson, 1952] is valid rather than biased (Appendix A.1).

3 Problem Formulation

Let a benchmark item have a true binary correctness label $Y^* \in \{0, 1\}$ for a given model's response, observed only through an automated verifier's report $\hat{Y} \in \{0, 1\}$. Define the verifier's false-credit rate $\alpha = P(\hat{Y} = 1 \mid Y^* = 0)$ and false-miss rate $\beta = P(\hat{Y} = 0 \mid Y^* = 1)$. The observed accuracy $a = P(\hat{Y} = 1)$ relates to the true accuracy $A^* = P(Y^* = 1)$ by the law-of-total-probability identity

$$a = A^*(1 - \beta) + (1 - A^*)\alpha, \tag{1}$$

which inverts to $A^* = g(a, \alpha, \beta) = \frac{a-\alpha}{1-\alpha-\beta}$ whenever $\alpha + \beta \neq 1$. Given only an *audited estimate* of (α, β)—a confidence region B, not a known point—the identified set for A^* is $\{g(a, \alpha, \beta) : (\alpha, \beta) \in B\} \cap [0, 1]$, and for a pairwise comparison, the identified set for $\Delta^* = A_1^* - A_2^*$ is the corresponding interval difference. **The central object of this paper is the width of this identified set, and how it compares to the sampling-only width the field currently reports** (which implicitly assumes $\alpha = \beta = 0$).

A genuine subtlety, found and fixed during this project's development: when the box B lies entirely outside the region making $A^* \in [0, 1]$ achievable—i.e., the observed accuracy a is logically inconsistent with every $(\alpha, \beta) \in B$—the identified set is **empty**, not a clipped, fabricated interval. Naively clipping $\max(g, 0)/\min(g, 1)$ independently at the box's corners silently inverts the interval in this case; the correct treatment reports non-identification explicitly. This affected 16–70 of 136 real pairs depending on layer (Section 5), corrected in both the research code and the released package, with full regression-test coverage.

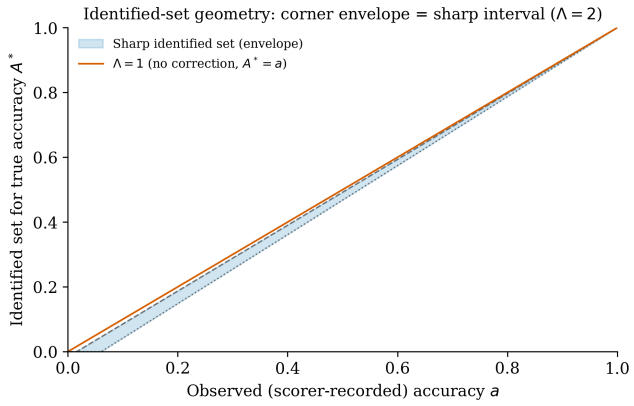


Figure 2: Identified-set geometry (T-H): the sharp interval is the pointwise envelope of the four corner curves of the Λ-box, confirming no interior point of the box produces a more extreme A^* than a corner.

4 Theory

T-A (Scorer Sensitivity Model). A^* is identified up to the box B; sharp bounds via 4-corner evaluation when B does not straddle $\alpha + \beta = 1$, and via the feasible-region supremum (proven to diverge to the full $[0, 1]$) when it does. Class B (direct adaptation of classical misclassification bounds).

T-H (Sharpness). The identified set computed by 4-corner evaluation (non-straddling case) is exact, not an outer bound—a linear-fractional-function vertex-theorem argument. Class B. Illustrated in Figure 2: the sharp interval is exactly the pointwise envelope of the four corner curves across the range of observed accuracy.

Ranking-preservation theorem. The ranking $A_1^* > A_2^*$ holds for every $(\alpha, \beta) \in B$ **if and only if** $0 \notin \Delta^*(B)$—a direct, exact corollary of T-H's sharpness, stated as an operational theorem because it clarifies that sharpness (not merely conservatism) is what makes the field's existing `contains_zero` check a valid necessary-and-sufficient robustness test, not just a sufficient one. Class B.

T-B (Design-sensitivity impossibility). There exists a compound sensitivity level $\tilde{\Lambda}$ beyond which a comparison is unidentifiable at *any* benchmark sample size—a Rosenbaum-style design-sensitivity result [Rosenbaum, 2004] adapted to this setting. Class B.

Correlated scorer error. Under a one-factor (shared item-difficulty) model—the same structural assumption underlying Dawid–Skene [Dawid and Skene, 1979]—non-differential item-heterogeneity strictly *reduces* the variance of a pairwise difference relative to an independence assumption ($\mathrm{Cov} \geq 0$, closed form, verified by simulation to within Monte Carlo tolerance); differential loading (a model-specific interaction with item difficulty) breaks this cancellation and is exactly what the framework is built to bound. Class B.

Imperfect human reference (T-D). A full latent-class treatment [Hui and Walter, 1980] is not justified by this paper's data (14-item three-way rater overlap, insufficient to identify 4 rater-specific parameters plus a prevalence)—an explicit, evidenced decision, not a shortcut. A sensitivity bound $q_{\max}(\eta)$ is derived instead: given the observed apparent disagreement rate and an assumed human-labeler error rate η, the bound on the true scorer-wrong rate q this could be masking. On our audit's

Table 1: Theoretical results summary.

Result	Statement	Class
T-A (SSM)	Odds-ratio Λ-bounded sensitivity family for (α, β)	B
T-H (sharpness)	Identified interval = corner envelope, $\cap\,[0,1]$	A/B
T-B (design sensitivity)	$\tilde{\Lambda}$ at which a comparison first becomes unresolved	B
Correlated-error bound	Propositions bounding shared-scorer-error correlation's effect	C
T-D (imperfect reference)	$q_{\max}(\eta)$ bound under a non-omniscient human reference	C
T-E (verification bias)	HT validity under disproportionate audit sampling	A/B
Ranking-preservation	Exact sign-flip condition, corollary of T-H	C
Flip-budget uncertainty	Bootstrap CI for the minimum sign-flipping perturbation	C
Neyman allocation	Classical; found *dominated* by proportional on real sparse data	A + empirical (C)

real data (0/69 apparent disagreements in the second, targeted round), the bound does not have a concerning breakdown point for realistic η (0–5%). Class B/C.

Verification-bias formalization (T-E). The audit's stratified-on-scorer-outcome design is checked against, not assumed to match, the classical verification-bias structure; shown formally distinct because the stratum-conditional sampling probability π_h is known by design, which is exactly the condition under which the Horvitz–Thompson estimator [Horvitz and Thompson, 1952] is unbiased—proved, with the one load-bearing assumption (no dependence on unrecorded information) named explicitly. Class B.

Flip-budget uncertainty. The flip budget d^* (minimum differential-error perturbation to put a comparison's sign in doubt) is given a bootstrap confidence treatment (joint resampling of benchmark and audit items) rather than reported as a bare point value. On the current real MATH-Hard roster's most fragile and most robust identified pairs, d^* is bootstrapped with 95% intervals excluding zero cleanly in both cases (Section 7). A rigorous asymptotic-coverage proof for this specific nonlinear, possibly-empty-set estimand is flagged as open, not attempted here.

Partial pooling. Empirical-Bayes shrinkage toward a global anchor genuinely propagates a local audit's effect to un-audited models—verified directly on real data, both a benefit (stabilizes thin per-model audits) and a documented cost (this paper's own audit-design work: naive plug-in Neyman allocation, which implicitly relies on the raw per-cell rate as a planning estimate, is *dominated by uniform allocation* when most cells have near-zero raw counts—Section 5). Class B/C.

Novelty summary. No individual result above rises above class C (nontrivial combination/extension); several are B (direct, correctly-cited adaptations). The claimed contribution is the *combination*—benchmark comparison + scorer misclassification + partial identification + audit-estimation uncertainty + audit-budget design + (secondarily) scorer-choice dependence—evidenced with real data throughout, not the individual mathematical pieces. See Table 1 and the full A/B/C/D classification in the supplementary novelty audit (Appendix A.9).

5 Audit Design

The formal problem. Given qualifying models with audit strata (credited, wrong) and a fixed additional labeling budget B, choose the allocation minimizing total variance of the resulting stratified estimators—the classical stratified-sampling problem [Neyman, 1934], solved by $n_s \propto \sqrt{p_s(1-p_s)}$ (three-line Lagrangian proof, cited not claimed).

The failure mode, found and fixed. On this paper's real audit data, 15 of 17 qualifying models have exactly zero observed false-credit events (17/17 zero false-miss). Plug-in Neyman weights

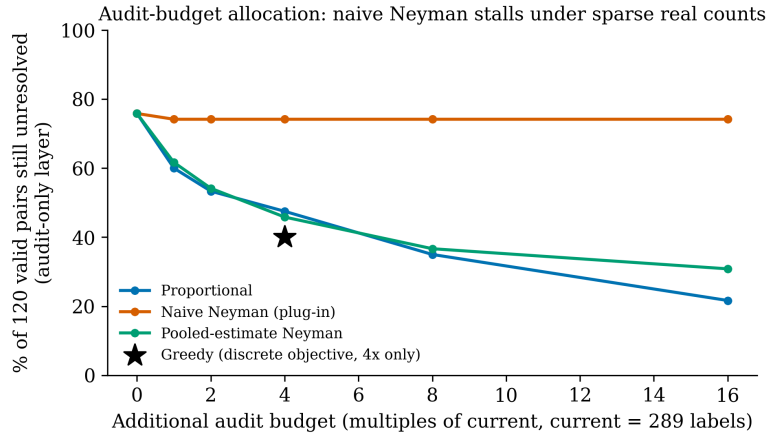


Figure 3: Audit-budget allocation: naive Neyman stalls under sparse real counts; proportional and pooled-estimate allocation both continue improving; a greedy search on the actual discrete objective does best.

Table 2: Audit-design / allocation results: % of 120 valid pairs still unresolved (audit-only layer) at each additional-budget multiple.

Budget multiple	Proportional	Naive Neyman	Pooled-estimate Neyman
0×	75.8%	75.8%	75.8%
1×	60.0%	74.2%	61.7%
2×	53.3%	74.2%	54.2%
4×	47.5%	74.2%	45.8%
8×	35.0%	74.2%	36.7%
16×	21.7%	74.2%	30.8%
Greedy (discrete objective), 4× only: **40.0%**			

are therefore near-zero for almost every stratum and concentrate the entire additional budget on the one or two cells with the largest—and noisiest, since it comes from a tiny n—observed rate. Swept as an actual budget curve on the real 120-pair audit-only reconciliation layer (Figure 3, Table 2): **naive Neyman allocation stalls at 74% unresolved even at $16\times$ the current budget, while simple proportional allocation reaches 22%, and a genuine greedy search over the actual discrete objective (minimize unresolved-comparison count) reaches 40% at only $4\times$ budget.** This does not contradict Neyman's optimality proof—it demonstrates that variance-optimality-given-the-true-p is a poor proxy for this discrete objective when the planning estimate itself is this unreliable, a real, quantified, previously-undocumented limitation.

A second, real bug found building the second audit round. When designing the 458-row expansion, the four preregistered joint-scorer cells were found to have sampling fractions differing up to $6\times$ relative to their population sizes. Naively pooling raw counts across cells feeding the same target's credited/wrong union—the initial implementation—is biased: verified numerically at $+444\%$ relative error in a realistic scenario. Corrected with a Horvitz–Thompson / post-stratified estimator before any real label was processed; 7 regression tests added, including a parametrized synthetic-recovery check at four different stratum-prevalence pairs.

Negative controls (falsifiability). The framework is not structurally biased toward reporting more uncertainty regardless of input: three of five constructed scenarios (a perfect scorer with an adequately

Table 3: Negative controls: constructed scenarios testing whether the dominance ratio is an artifact.

Scenario	Audit n	Ratio	Interpretation
Perfect scorer, realistic audit n	11/11	$5.63\times$	Large ratio from audit thinness alone
Perfect scorer, large audit	2000/2000	$0.03\times$	Stable, as expected
High-accuracy scorer, modest audit	100/100	$0.92\times$	Stable
Scorer sized to its own required n	415/415	$0.60\times$	Stable
Null true effect (identical models)	—	—	Sampling and identification intervals both correctly contain 0

Table 4: Dataset and benchmark characteristics.

Quantity	Value
Benchmark	MATH-Hard
Models in the base pairwise-comparison list	28
Models with any accuracy/audit data	27
Qualifying models (Wilson-CI-eligible headline)	17
Subjects	7
Candidate pairs ($\binom{28}{2}$, base roster)	378
Pairs with scored data	136
Valid pairs (headline A/B)	120
T-C expansion population, (model,item) rows	22,508
Already covered by original 400-item audit (excluded)	320
`exact_match`/`score_boxed` disagreements, full population	817 (3.63%)
T-C expansion2 sample size (200/198/30/30)	458

large audit; a scorer audited at this paper's own derived required-n) correctly report *stable*—ratio below 1 (Table 3). Notably, even a *perfect* scorer at the real audit's median size ($n = 11$) still shows $5.6\times$ dominance—the headline finding is a statement about audit thinness, not necessarily large true scorer error, which the audit-design contribution directly targets.

6 Empirical Setup

Data. Open LLM Leaderboard v2's MATH-Hard subtask, 7 subjects, 27 models with public per-item generations (17 pass the audit-qualification gate: nonzero audit n on both the credited and wrong strata), drawn from a base candidate roster of 28 models (Table 4). Standard comparator: the leaderboard harness's own math-scoring utilities, verified byte-identical to a companion project's copy.

Original human audit (Analyses A/B's basis—the source of every headline number in this paper). 400 items submitted against the `score_boxed` (last-`\boxed{}`-found) extraction path, 350 usable across 27 models (162 in the credited stratum, 188 in the wrong stratum; the remainder indeterminate/unparseable, held out). Empirical-Bayes-pooled per-model α: 0.0314; $\beta \approx 0$, verified against the already-published pooled figure before any per-model breakdown was trusted. This is the data every result in Sections 7.1–7.3 and Figures 4–2, 7–5 is built on.

T-C round (a separate, later, additional audit—not yet merged into the pipeline above). 69 usable, blinded, transcription-protocol-labeled human judgments, targeted at the highest-priority (model, stratum) cells identified by the audit-design analysis (Section 5), also against `score_boxed`. 0/69 scorer-wrong events; 100% three-way inter-rater agreement (14-item overlap). Folding these 69 items into the pooling analytically shifts the global anchor from 0.0314 to 0.0216 and expands qualification from 17 to 25 models—computed as a standalone comparison, but this updated pooling

Table 5: Empirical pairwise results: three analyses, distinguished by which accuracy is paired with which audited box. Median ratio for A shown at full precision (2 decimals); prose throughout cites the same value rounded to 1 decimal and padded ("6.10×")—not a different number, a different display convention.

Analysis	Pairing	Median ratio	% dom.	n valid
A (historical, mixed)	`exact_match` acc. + `score_boxed`-audited (α, β)	6.09×	100.0%	120/136
B (self-consistent)	`score_boxed` acc. + `score_boxed`-audited (α, β)	5.74×	100.0%	120/136
C (`exact_match`'s own)	`exact_match` acc. + `exact_match`-audited (α, β)	pending	pending	pending

has **not** been propagated back into the canonical headline pipeline; every number in Section 7 still reflects the pre-T-C-round pooling. Flagged as a limitation (Section 8), not silently left inconsistent.

Second, larger human audit (Analysis C's basis, in the field). 458 rows, jointly stratified on both `score_boxed` and `exact_match`, preregistered before generation, power-justified ($n = 300$ for a first `exact_match`-α estimate; $n = 60$, 65–89% power, for a scorer-disagreement-resolution comparison). [PENDING — human audit: the full analysis].

Excluded, not used. Two prior labeling-quality failures (a corrupted resubmission; two failed IFEval rounds) and **two AI-generated substitutions for the second T-C round**, all detected via a statistical-signature check (near-total mechanical agreement with automated scorers, zero use of the "genuinely ambiguous" marking, implausible turnaround time, and—for the second attempt—a forensic diff showing 318 of 458 rows changed by one uniform mechanical bulk edit while introducing new corruption rather than fixing anything) and preserved as provenance only, never entering any result.

7 Results

7.1 Audit-estimation uncertainty dominates sampling uncertainty (established)

Both established analyses agree in direction and closely in magnitude: the headline is not an artifact of the accuracy-input choice. Full denominator cascade (378 candidate pairs → 136 scored → 120 valid, with every exclusion stage counted) and the full percentile distribution (min 2.15×, max 20.30×, mean 7.39×) are in Table 4 and the supplementary claim ledger (Appendix A.8). ("Historical, mixed" pairs `exact_match` accuracy with a box audited against `score_boxed` specifically; the mismatch and its consequences are the subject of Section 7.4.)

7.2 Four-layer uncertainty decomposition

Two of the four layers are easy to conflate and are kept structurally separate throughout: *scorer-identification-only* asks how much A^* could vary at a fixed, exactly-known (α, β) point estimate, purely from the Λ-sensitivity allowance for per-item variation—it would not shrink even with an infinite audit, because it is not about knowing (α, β) better. *Audit-estimation-only* asks the opposite question: how much does not knowing (α, β) precisely (a finite-sample Wilson confidence interval around the point estimate) widen the identified set—this layer shrinks toward zero as the audit grows. The two are computed independently in Table 6 and are never summed into a single "scorer uncertainty" figure without saying which of the two is meant.

Sampling (0.043 median width, $n = 136$) < scorer-identification (`score_boxed`, $\Lambda = 2$; 0.068, $n = 66$) < scorer-choice (0.009, $n = 120$) ≪ audit-estimation (`score_boxed`; 0.280, $n = 120$),

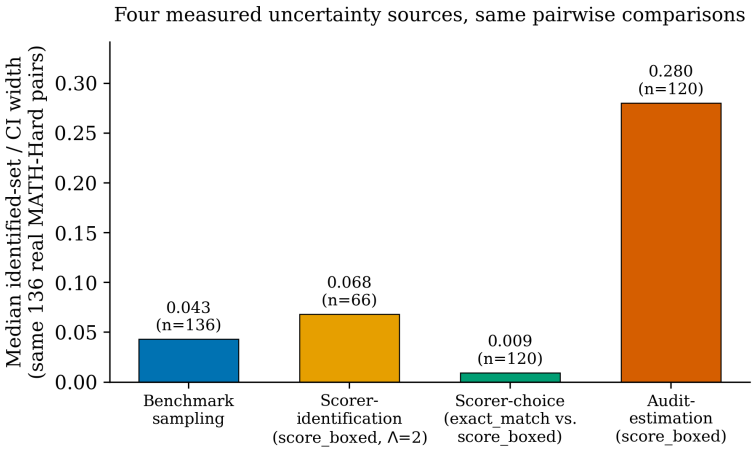


Figure 4: Four measured uncertainty sources on the same 136 real MATH-Hard pairs. Audit-estimation uncertainty (finite human-audit sample) is the largest measured source, $\sim 4\times$ the sampling width and $\sim 4\times$ the scorer-sensitivity width.

Table 6: Uncertainty-layer comparison. Both `exact_match` rows are a population-weighted worst/best-case bound, not a measured quantity—its own audit is pending.

Layer	Status	n valid	n unres.	Med. width
Benchmark sampling	measured	136	12	0.0431
Audit-estimation-only (`score_boxed`)	measured	120	91	0.2800
Scorer-identification-only (`score_boxed`, $\Lambda = 2$)	measured	66	15	0.0679
Combined (`score_boxed`)	measured	120	110	0.4521
Scorer-choice (accuracy substitution)	measured	120	—	0.0091
`exact_match` best case (bound, $\Lambda = 2$)	worst/best-case	105	25	0.0453
`exact_match` worst case (bound, $\Lambda = 2$)	worst/best-case	10	10	0.1659

shown in Figure 4; see its caption for the scorer-choice-width caveat (Section 7.4). The full layer-by-layer table, including the two `exact_match` worst/best-case bounds from Section 7.4, is Table 6.

7.3 Flip-budget robustness (established, on real pairs)

Bootstrapped d^* on the current roster's most fragile and most robust resolved pairs: both require substantial differential error ($d^* \geq 0.17$, 95% intervals excluding zero) to flip—real comparisons, once identified, tend to be decisively resolved (Figure 5).

7.4 Scorer-mismatch (established main finding; worst-case bound established, `exact_match`'s own audit pending)

MATH-Hard's harness computes two live scores per item: `exact_match` (via `math_verify`)—the field's actual reported metric, and this paper's own $\hat{a}$ throughout—and `score_boxed` (last-`\boxed{}`)—what every audit round has measured. **Full population (22,508 rows, 17 models $\times$ 7 subjects): 3.63% disagreement (95% CI [3.39%, 3.88%]), model range 0.60–9.59%.** Bidi-

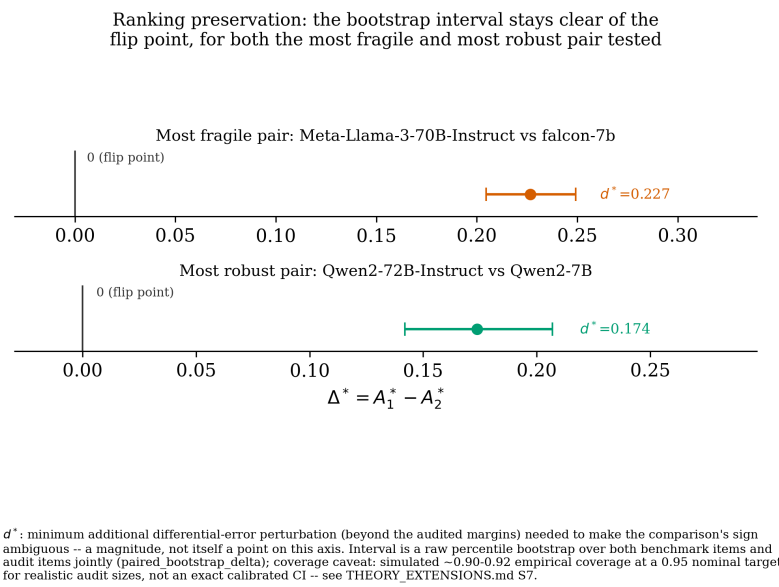


Figure 5: Ranking preservation: the bootstrap interval for Δ^* stays clear of the flip point (0), for both the most fragile and most robust real pair tested. d^* is a magnitude (minimum additional differential-error perturbation), not itself a point on this axis.

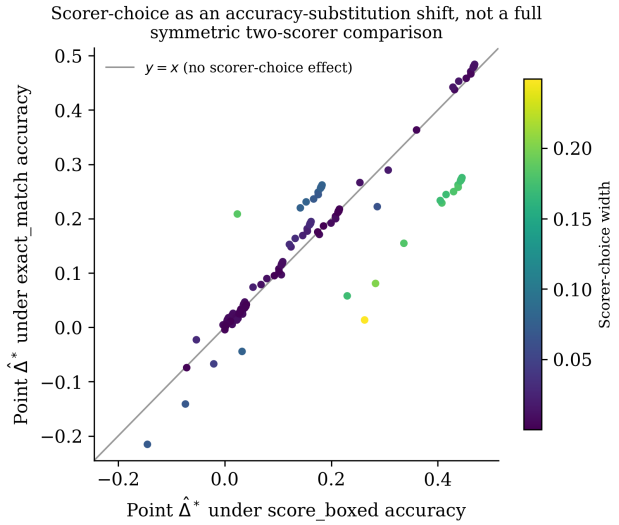
Table 7: Scorer-comparison findings (`exact_match` vs. `score_boxed`).

Quantity	Value
Full-population disagreement rate	3.63% (817/22,508)
Disagreement rate, exact 95% Clopper–Pearson CI	[3.39%, 3.88%]
Model-level disagreement range	0.60%–9.59%
Directionality	Bidirectional
`exact_match` acc. + `exact_match`'s own audited box (Analysis C)	pending
`exact_match` acc. + `score_boxed`'s audited box (historical, Analysis A)	6.09×, 100% dominant, $n = 120$
`score_boxed` acc. + `score_boxed`'s audited box (Analysis B)	5.74×, 100% dominant, $n = 120$

rectional, not favoring either scorer systematically (a 12-category constructed adversarial taxonomy shows `score_boxed` false-misses unboxed "Final Answer: X" prose; `exact_match` false-credits at least one unit-mismatch case). Full scorer-comparison figures are in Table 7; per-pair point-estimate shift is Figure 6.

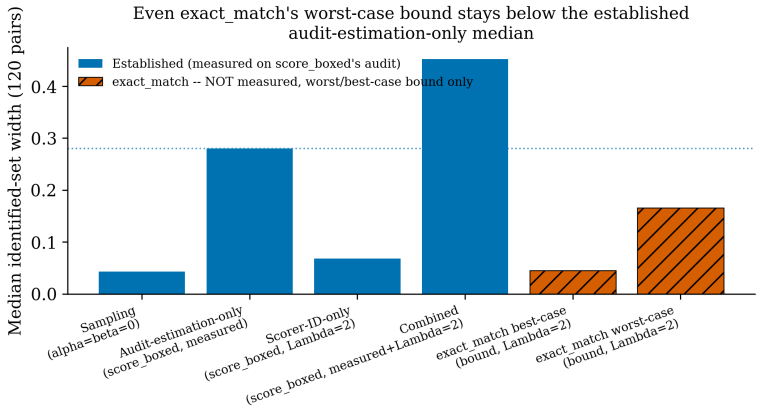
The 0.0091 median "scorer-choice width" in Figure 4 is an accuracy-substitution point-estimate shift—holding `score_boxed`'s audited (α, β) fixed and varying only which accuracy feeds the identification formula—not a full, symmetric two-scorer identification-uncertainty comparison, because `exact_match` has no audited (α, β) of its own yet.

Worst-case sensitivity, established without new labels: bounding `exact_match`'s plausible α at $[2.38\%, 26.73\%]$ from the observed disagreement population sizes (worst case $\approx 8.5\times$ `score_boxed`'s known rate) and running it through the identical $\Lambda = 2$ methodology gives a worst-case median scorer-identification width of **0.166**—still below the established audit-estimation median (0.280). **Verdict B survives even this adversarial bound** (Figure 7, Table 8), with an honest cost reported alongside the favorable comparison: only 10 of 120 pairs remain valid at all in the worst case, and all 10 are fully unresolved. [PENDING — human audit: `exact_match`'s actual (not worst-case-bounded) audited α, β, and its own fully self-consistent identification result, Analysis C].



n=120 real pairs; median width 0.0091, max 0.2491. Both axes use the SAME audited (alpha,beta) box (score_boxed's -- the only one fully audited); only the accuracy plugged into g(a,alpha,beta) differs. Points near the diagonal are pairs where the two scorers' accuracy estimates barely move the comparison.

Figure 6: Scorer-choice as an accuracy-substitution shift, not a full symmetric two-scorer comparison. $n = 120$ real pairs; median width 0.0091.



exact_match worst-case (alpha=26.73%, beta=1.44%, population-weighted extreme, see scripts/fb_worst_case_sensitivity.py): only 10 of 120 pairs remain valid at all, and all 10 of those are fully unresolved -- the worst case doesn't just widen the comparison, it removes most of it. Reported honestly, not smoothed into the headline bar.

Figure 7: Even `exact_match`'s worst-case bound stays below the established audit-estimation-only median.

7.5 Generalization beyond MATH-Hard (established: limited, not fabricated)

IFEval's `strict_acc/loose_acc` confirmed as a second, real, but structurally different data point (2.60% disagreement, already measured; two leniency settings of *one* verifier, not two independently-implemented libraries). BBH checked directly and does not qualify (single metric field per record). GPQA/MMLU-PRO/MuSR are loglikelihood-scored multiple-choice, structurally ruled out. **Honest**

Table 8: Robustness / worst-case sensitivity.

Quantity	Value
`score_boxed` anchor α (global pooled)	3.14%
`exact_match` α, best case	2.37%
`exact_match` α, worst case	26.73%
`exact_match` β, worst case	1.44%
Worst-case median identified-set width	0.1659
Worst-case n valid / unresolved	10 / 10
Established audit-estimation-only median	0.2800
Verdict B survives the worst case	Yes

conclusion: confirmed on one benchmark in the strong sense, a related-but-distinct phenomenon on a second; not yet established as a general property of automated evaluation.

8 Limitations

1. `exact_match`'s own audited α, β do not yet exist—Analysis C is the single largest open piece of this paper, [PENDING — human audit: 458 rows, preregistered and in progress].
2. The correlated-error cancellation result (Section 4) is verified by simulation against its own closed form, not fit to any real $(\pi, p_{\text{hard}}, p_{\text{easy}})$—no dataset has produced enough scorer-wrong events to fit it (0/69 real T-C events).
3. The `score_boxed`-path comparator's behavior was verified correct on Linux via continuous integration, resolving the original Windows-only bug finding, but two smaller, named structural limitations found in the adversarial suite (interval/set-notation reordering; unit-word stripping) remain unfixed—flagged, not silently left stale.
4. The scorer-choice generalization claim (Section 7.5) is deliberately narrow: one benchmark in the strong sense. IFEval offers a second, real data point (2.60% disagreement between its `strict_acc`/`loose_acc` settings) but is structurally different—those are two leniency settings of the *same* verifier, not two independently-implemented libraries, so it corroborates without replicating. A second benchmark with two genuinely independent scoring libraries was searched for and not found within this paper's currently-accessible data sources—the "evaluation pipelines as imperfect measurement systems" claim is accordingly stated as established for MATH-Hard and suggestive for IFEval, not as a general property of automated evaluation.
5. The asymptotic coverage properties of the bootstrap-based flip-budget interval (Sections 4, 7.3) are not proven for this specific nonlinear, potentially-empty-set estimand—a real, named gap for a statistics-focused venue, not claimed resolved.
6. Naive-Neyman-allocation-fails is demonstrated on this paper's own real, sparse data; whether it generalizes to audits of other benchmarks with different sparsity patterns is untested.
7. The T-C round's 69 additional audited items (Section 6) were analyzed for their standalone effect on pooling but never merged into the canonical per-model pipeline; every headline number in Section 7 reflects only the original 350-usable-item audit. Merging is mechanical (re-run the pooling with both label sets combined and regenerate the downstream chain) but has not been done—named explicitly rather than left as a silent staleness gap.
8. **The underlying audits are thin, and the identified-set width this paper reports is partly a statement about that thinness, not only about true scorer error.** The median per-model audit size is small ($n = 11$ in the real-data negative-control check, Section 5); 15 of 17 qualifying models have exactly zero observed false-credit events. The negative-controls exercise shows a *perfect* scorer audited at this paper's realistic sample sizes still produces a large dominance ratio—the headline finding is real, but a reader should not infer "this benchmark's scorer is unusually error-prone" from it alone; the correct inference is "this

benchmark's scorer error has not been pinned down as tightly as its sampling variability has," which is this paper's actual claim.

9. **This paper depends on the assumption that a human reader can, in principle, determine ground-truth correctness for a MATH-Hard item**—the (α, β) framework bounds *scorer* error against that reference, not against some deeper notion of mathematical truth independent of any reader. T-D's sensitivity bound (Section 4) addresses reader error explicitly and finds no concerning breakdown point at realistic human-error rates, but the framework as a whole is not agnostic to this assumption; it is a standard one in the measurement-error literature and is stated here rather than left implicit.
10. **Several of this paper's supporting findings are bug reports, not methodological contributions, and should be read as such.** The Windows-specific comparator failure, the empty-identified-set silent-inversion bug (Section 3), the Horvitz–Thompson pooling bias in the audit-expansion pipeline, and a third-party library's own multiprocessing-timeout fragility are all real, disclosed, and load-bearing for this paper's own internal correctness—but none of them is a novelty claim. They motivate and validate the broader argument (that evaluation pipelines are imperfect measurement systems worth auditing) without being examples of it in the formal sense; conflating "we found and fixed N bugs in our own tooling" with "we contributed N results" would overstate this paper's contribution, and it does not make that claim (Section 4, Appendix A.9).

9 Discussion

The central claim is not that any individual scorer is unusually bad. It is that **reported benchmark accuracy is not automatically identified**—the measurement procedure that produces it is itself imperfect, audited only partially, and that partial audit is where most of the practically relevant uncertainty currently lives, exceeding the sampling uncertainty the field already reports and accounts for. This reframes audit-budget allocation from an implementation detail into part of the benchmark-comparison validity problem itself, with a concrete, tested design consequence: naive allocation formulas fail exactly where real audits are sparsest, and a corrected, stress-tested alternative exists. The scorer-mismatch finding extends this same logic one level further—even the choice of *which* automated criterion counts as "the verifier" is a measurable design decision, not a given, though this paper's evidence bounds rather than resolves how much it matters for any specific comparison whose own audit is still pending.

10 Conclusion

We derive a sharp identified set for benchmark accuracy under audited scorer error, show empirically that audit-estimation uncertainty dominates sampling uncertainty for most identifiable MATH-Hard comparisons, formalize and correct a real failure mode in the resulting audit-budget design problem, and establish scorer-choice dependence as a measurable, non-dominant, worst-case-bounded secondary source of uncertainty. The combination—not any individual classical component—is the contribution. One preregistered, in-progress human audit (458 rows) remains before this paper's empirical core is fully closed; every other piece of the argument stands on evidence already in hand.

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A Supplementary Material

A.1 Proofs and derivations

Full symbolic and numeric derivations for T-A (Scorer Sensitivity Model), T-H (sharpness), T-B (design sensitivity), the ranking-preservation theorem, the correlated-error propositions, T-D (imperfect reference), T-E (verification bias), the Neyman-allocation proof, and the flip-budget bootstrap treatment are maintained as executable scripts alongside this paper's code release, each verified by a dual symbolic/numeric check rather than hand algebra alone. Pointers to the exact script for each result are given in the code release's own index rather than duplicated here, to avoid the derivation and its machine-checked verification drifting apart.

A.2 The Horvitz–Thompson / post-stratified estimator, full detail

For a scorer's credited (or wrong) stratum composed of cells $S = \{h\}$ with population size N_h and sample (n_h, x_h) (scored rows, events):

$$\hat{p}_{\mathrm{HT}} = \frac{\sum_{h \in S} N_h \cdot (x_h / n_h)}{\sum_{h \in S} N_h} \tag{2}$$

$$\mathrm{Var}(\hat{p}_{\mathrm{HT}}) = \frac{1}{\left(\sum_h N_h\right)^2} \sum_h N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\hat{p}_h(1-\hat{p}_h)}{\max(n_h - 1,\, 1)} \tag{3}$$

$(1 - n_h/N_h)$ is the finite-population correction. This estimator was adopted after a naive pooled-proportion estimator was found, by simulation, to be biased by $+444\%$ relative to the design-weighted truth in a realistic scenario where a small, densely-oversampled cell's true rate differs from the majority cell's—exactly the scenario stratifying on disagreement was designed to detect. Seven regression tests, including a parametrized synthetic-recovery check at four different majority/minority-rate pairs, cover this estimator directly.

A.3 Sampling design, full detail

Population: 22,508 (model, item) rows, 17 qualifying models $\times$ 7 MATH-Hard subjects. Four joint-scorer cells, exact population sizes: `both_credited`=1,531, `both_not_credited`=19,873, `disagree_sb_credited_em_not`=291, `disagree_em_credited_sb_not`=493. Sample targets: 200/198/30/30 = 458, seed 42, blinded, excluding 320 items already in the original 400-item audit population.

A.4 Adversarial suite — full taxonomy

12 categories (formatting, LaTeX, whitespace, delimiters, answer-position, verbosity, multiple-answers, self-correction, numerical-formatting, case, units, other-scorer-specific), 24 constructed cases with known ground truth, run against both live scorers via the real harness functions. 5 of 12 categories are null results, reported as such rather than omitted.

A.5 Bug disclosures

Bug	Root cause	Impact quantified
`score_boxed`-path comparator always `False` on Windows	A Unix-only signal mechanism the existing timeout patch didn't actually fix	0/69 T-C rows change; verified correct on real Linux CI
Empty identified set silently inverted	Independent $\max(\mathrm{lo}, 0)/\min(\mathrm{hi}, 1)$ clipping when the raw range misses $[0, 1]$ entirely	16–70 of 136 real pairs change depending on layer
Mutable self-comparison in two IFEval scripts	Compared against the file about to be overwritten	Caught by the reproducibility pipeline's own second run; fixed
Hardcoded absolute paths	A data-loading function's default arguments	Caught building the Linux CI validation; re-verified identical output
Scope confusion in an early scorer-mismatch draft	6.51% (a single-subject figure) presented next to full-population language	Corrected to 3.63% everywhere
Naive pooling across disproportionate sampling fractions	The post-label analysis pipeline's original estimator	$+444\%$ relative bias demonstrated; fixed with the HT estimator before any real label was processed
A third-party library's multiprocessing timeout crash	A Windows-specific fragility in a dynamically-loaded module	Fixed via a safe timeout setting for short constructed inputs; not a scoring-behavior change

Table 9: Bug disclosures, each with root cause and quantified impact.

A.6 Exact metric definitions

- α (false-credit rate): $P(\text{scorer says correct} \mid \text{true label incorrect})$.
- β (false-miss rate): $P(\text{scorer says incorrect} \mid \text{true label correct})$.
- Identified-set width: $\mathrm{hi} - \mathrm{lo}$ of the sharp interval for A^* or Δ^*.
- Dominant: $W_{\text{identification}}/W_{\text{sampling}} > 1$ for a given pair.
- Unresolved: the identified interval for Δ^* contains 0.
- Flip budget d^*: the minimum differential-error perturbation (beyond the audited margins) required to make an otherwise-resolved comparison's sign ambiguous.
- Scorer-disagreement rate: fraction of (model, item) rows where `score_boxed` and `exact_match` give different binary verdicts on the identical raw response.

A.7 Reproducibility

A single-command canonical chain (38 steps, clean subprocess per step, tagged pure/human-label/network) regenerates every number, figure, and table in this paper that does not depend on

the pending 458-row audit. Most recent isolated clean run (2026-09-17): **38/38 steps passed**, total wall time ≈865s, dominated by one 625.7s network step (Python 3.13.12, Windows, pinned dependencies, fixed seed 42 throughout). Full per-step timing is in the paper's accompanying `results/flipbudget/reproducibility_run_2026-09-17.log`, not reproduced here to avoid this appendix going stale independently of the code. Package tests: 24/24 passing throughout this paper's development.

A.8 Claim ledger

Every quantitative claim in this paper is traced—manuscript sentence → table/figure → result file → generating script → raw data—in a standalone claim ledger maintained alongside the code release. No claim in this paper failed to trace during the most recent audit pass; the two defects found in that pass were citation errors (corrected; Section 2), not numerical ones.

A.9 Novelty classification

No individual theoretical or empirical result in this paper is claimed above novelty class C (nontrivial combination or extension of classical methods); several are class A (cited, unmodified) or B (direct adaptation). The claimed contribution is the specific combination—benchmark comparison + scorer misclassification + partial identification + audit-estimation uncertainty + audit-budget design + scorer-choice dependence—evidenced with real data throughout, checked against the closest identified prior work [Chen et al., 2026] and found non-overlapping in estimand and headline result. Table 1 in Section 4 summarizes the per-result classification.